Phase 1 Strategy Report: Preprocessing and Feature Engineering Plan

Predicting Minimum Proficiency Risk in African Primary Education Using PASEC Survey Data

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Repository: `https://github.com/GentilleUwera/pasec-minimum-proficiency-risk`

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1 Purpose and scope

This report audits the raw PASEC2019 Grade 6 data and fixes every preprocessing decision before I write the production pipeline. The task is binary classification: from background information only, flag a Grade 6 student as likely below the PASEC minimum proficiency level in mathematics. Every figure in this report was computed on the real file (`PASEC2019_GRADE6_INT.dta`). Statistics that guide modelling choices (skewness, outlier bounds, associations, collinearity) were computed on the training partition only, after the split described in Section 5, so the test set has not influenced any decision.

2 Dataset and problem audit

2.1 Initial dimensions

The raw matrix is **N = 62,934 rows by P = 749 columns**. Each row is one Grade 6 student. The students come from 14 countries and **3,298 schools**. The school code `ID_ECOLE` restarts in every country (1,173 distinct codes), so a school is only uniquely identified by country plus `ID_ECOLE`. `ID_TOT` is unique for every row, so there are no duplicate students. The file merges three sources: the student test and questionnaire, the questionnaire of the class teacher, and the questionnaire of the school principal. Teacher and principal answers are therefore repeated for every student in the same class or school, which matters for the split (Section 5).

2.2 Variable inventory

I classified every column using the PASEC2019 data manual and the value-label sets in the AFLEARN English labels file. A column attached to an ordered label set (for example Always, Often, Sometimes, Never) is ordinal; an unordered set (Yes/No, school status, water source) is nominal; numeric counts with a "[min-max]" range (ages, days, numbers of classrooms) are discrete; indices, scores, money amounts and weights are continuous; date parts are datetime. Test items are multiple-choice responses A to D and are nominal. There are no free-text columns.

Three findings from the inventory shape the plan:

- **Redundant and empty columns.** Birth year `qe61y` equals 2019 minus age `qe62` for 100% of students, so it carries no extra information. The two course-date columns `qd26` and `qd27` are 100% empty.
- **Inconsistent coding.** The principal and teacher education and diploma variables (`qd11`, `qd12`, `qm3`, `qm4`) are coded per country, and their codes are not in rank order. I do not use them.
- **Columns that would leak the answer.** The 182 test items and 10 plausible values contain the outcome or the responses it is built from (Section 5.5).

Selected features. I map each background concept from my proposal to one verified column:

Country is an addition to the proposal. Pooling 14 education systems was flagged as a challenge in Assignment 1, and country is the strongest single predictor in the training data (Cramér's V = 0.394), so leaving it out would force the other features to absorb system differences. Gender (`qe63`) is kept out of the feature set and used only for the fairness audit. It shows no association with the label in the training data (Cramér's V = 0.000, p = 0.93). Two points remain unverified: the

Table 1: Variable types by block (all 749 columns). Every column is listed with its type in Appendix A.

Block	Continuous	Discrete	Nominal	Ordinal	Datetime	Total
Identifiers and context (IDs, country, test language)	0	0	8	0	0	8
Test items (f1 to f98, m1 to m84)	0	0	182	0	0	182
Plausible values (reading and maths)	10	0	0	0	0	10
Student questionnaire (qe6)	0	1	78	13	3	95
Teacher questionnaire (qm)	0	20	93	31	0	144
Principal questionnaire (qd)	4	26	151	21	8	210
Derived indices (ses, INDICE_*)	6	0	0	0	0	6
Survey design (weights, jackknife zones)	92	0	2	0	0	94
Total	112	47	514	65	11	749

Table 2: Selected features and their types.

Concept (proposal)	Column	Content	Type
Household assets	`ses`	PASEC socio-economic index built from household possessions (mean 50, SD 10)	Continuous
Parental literacy	`qe621a`, `qe621b`	Mother can read; father can read (Yes/No)	Nominal (binary)
Grade repetition	`qe68`	Ever repeated a grade (Yes/No)	Nominal (binary)
Age-for-grade	`qe62`	Student age in years	Discrete
Teacher qualification	`qm5`	Length of initial teacher training, 6 ordered levels, same coding in all countries	Ordinal
Class size	`qm22`	Students enrolled in the class	Discrete
School location	`qd31`	City, suburb, large village, small village	Ordinal
Home language	`qe620`	Speaks French at home: always, often, sometimes, never	Ordinal
Country (added)	`ID_PAYS`	14 countries; captures system-level differences when pooling	Nominal

urban/rural grouping of `qd31` (city and suburb versus villages) is my own, since PASEC's urban variable is not in the file; and I could not confirm whether `qe620` was adapted for the 2,130 students tested in English (Cameroon) or Arabic (Chad).

2.3 Baseline missingness

PASEC codes missing data in two ways. Some cells are blank in the file, mostly when a teacher or principal questionnaire could not be linked to the student. Other cells hold a sentinel code: Invalid, Missing, Not applicable, Missing by design, or Not reached. The codes depend on the variable. For example, 6 means "Invalid" for Yes/No items but "River" for the water-source question, and 9 is a real diploma code in `qm4`. I therefore convert codes to missing variable by variable, using a map of 627 variables built from the AFLEARN label sets. The AFLEARN file leaves nine variables uncovered (`qd11`, `qd12`, `qm1`, `qm3`, `qm4`, `se8`, `qd73`, `qd75a` to `qd75c`). For these I used the codes listed in the PASEC manual annexes (`qm1`: 6, 9; `qm4` and `qd12`: 99; `qd75a` to `qd75c`: 9999999996, 9999999997, 9999999999). A few codes appear in the file but are not listed in the manual for that variable: 96 and 99 in `qd11` and `qm3`, 96 in `qm4` and `qd12`, 9999 in `se8` (its real values stop at 365), and the 10-digit codes in `qd73` (the manual lists 96, 97 and 99, which never occur, while the sibling variables `qd75a` to `qd75c` use the 10-digit codes). I treat these as missing and flag them as inferred. They cover 39,206 cells (0.08% of the matrix), all in columns that are not model features.

- **Overall:** 25.69% of all cells are missing, or 17.0% once the 182 test items are set aside. Most of the gap comes from the booklet design: each student sees only some test items.
- **Complete versus incomplete rows:** 0 of 62,934 rows are complete; every row has at least one missing value, for the same booklet reason. On the 10 selected feature columns, 41,661 rows (66.2%) are complete and 21,273 (33.8%) have at least one missing value.
- **Per column:** 110 columns have no missing values, 81 have up to 5%, 230 have 5 to 20%, 85 have 20 to 50%, and 243 have more than 50% (mostly test items and skip-logic questions). The count and percentage for every column are listed in Appendix A.
- **Selected features:** missingness ranges from 0.3% (age) to 14.0% (class size). See Figure 1.

Table 3: Missing cells in the full raw matrix (62,934 × 749 = 47,137,566 cells).

Source of missingness	Cells	% of all cells
Blank in file (questionnaire not linked, index not computed)	1,288,006	2.73
Missing (question not answered)	3,102,232	6.58
Not applicable (question skipped by design of the questionnaire)	1,710,706	3.63
Invalid (several answers ticked)	162,616	0.34
Missing by design (test item not in the student's booklet)	5,819,246	12.35
Not reached (test item at the end of the booklet)	26,197	0.06
Total	12,109,003	25.69

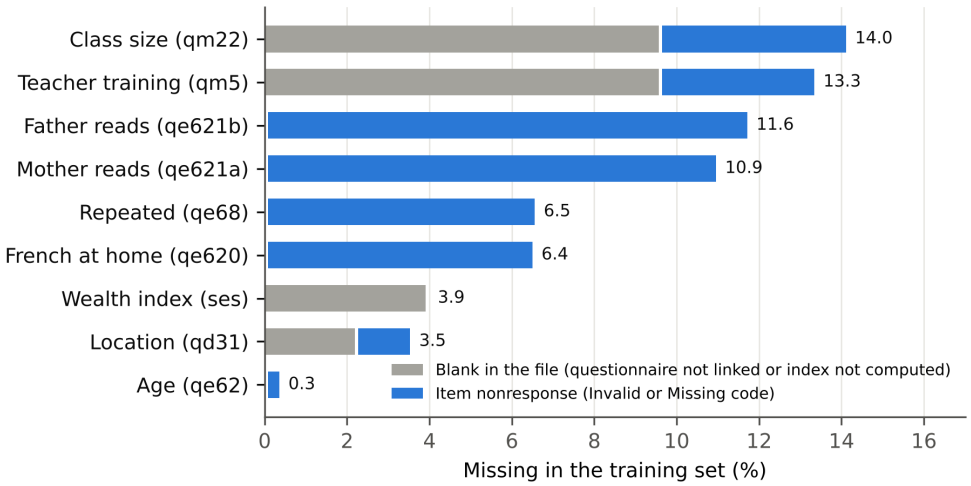


Figure 1: Missingness of the selected features in the training set, split into blank cells and item nonresponse. Teacher variables share a 9.6% block of students whose teacher questionnaire is not linked.

2.4 Target label (Y)

Task type: binary classification. Y = 1 if the student is below the PASEC minimum proficiency level in mathematics, 0 otherwise.

Cutoff. The PASEC2019 manual (Table 5.17) sets the end-of-primary mathematics boundary at 521.46. The AFLEARN guide identifies this boundary as the PASEC minimum proficiency level. AFLEARN also notes that the SDG 4.1.1b minimum for Grade 6 sits one level higher (609.64). At that cutoff 88.7% of students would be positive, which leaves almost nothing to predict, so I use the PASEC minimum and name it as such.

Plausible values. PASEC does not give one maths score per student. It gives five plausible values, five equally likely draws of the student's ability. The manual requires every statistic to be computed on each plausible value and then averaged. For a per-student label I treat the five values like five annotators:

- **Training label:** Y = 1 when at least 3 of the 5 values are at or below 521.46 (majority vote).
- **Agreement:** the five per-value labels agree with Fleiss' kappa = 0.769 (substantial agreement). For 22.2% of students the five values do not all fall on the same side of the cutoff, so these students are genuinely borderline.
- **Evaluation:** test metrics will also be computed against each of the five per-value labels and averaged, following PASEC's rule.

The plausible values were generated with background variables in PASEC's scaling model, so part of the link between my features and the label is built in. I report this as a limitation, not as leakage.

Class balance. The full dataset has 39,685 positives (63.1%) and 23,249 negatives (36.9%), an imbalance ratio of 1.71:1. The training set matches this (31,751 vs 18,596). With PASEC's student weight `rwgt0`, the population share below the minimum is 69.1%. The rate varies strongly by country, from 29.9% (Gabon) to 87.6% (Chad) (Figure 2).

Mitigation. The imbalance is mild and the class of interest is the majority, so I do not resample:

- **No SMOTE.** SMOTE would invent students by interpolating between coded survey answers, and a value halfway between "Yes" and "No" has no meaning.
- **No random undersampling.** It would throw away real students.

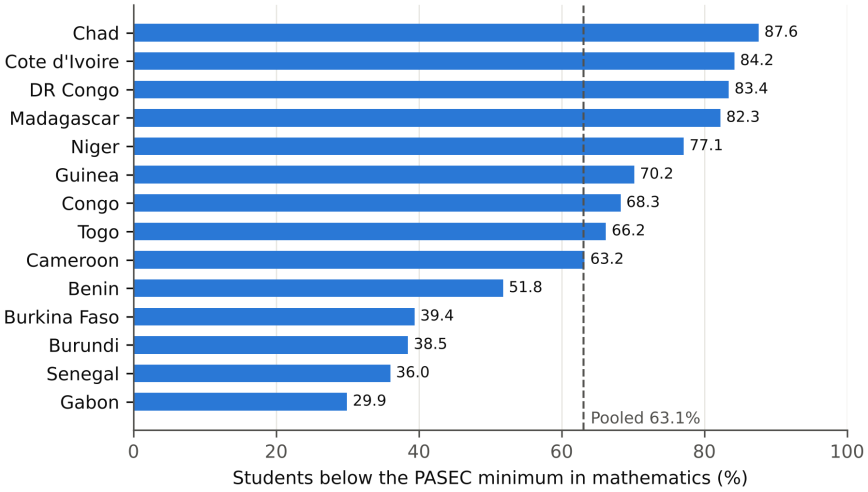


Figure 2: Share of students below the PASEC minimum in mathematics, by country (unweighted, full dataset).

- **Balanced class weights instead** (`class_weight='balanced'` for logistic regression, the equivalent sample weights for XGBoost), so the model cannot win by predicting the majority class.
- **Stratified splitting** (Section 5).
- **A tuned decision threshold.** The threshold is chosen on out-of-fold predictions to reach the recall target, instead of the default 0.5.

Evaluation consequence. Because 63% of students are positive, a classifier that flags everyone scores recall 1.00 and F1 0.773, and so meets my Assignment 1 targets (recall ≥ 0.75, F1 ≥ 0.70) without learning anything. I therefore add this all-positive classifier as a second baseline next to the wealth-quintile baseline, and report balanced accuracy (0.50 for the all-positive classifier) and PR-AUC with every model. The wealth-quintile comparison (+10 F1 points) remains the main test of added value.

3 Cleaning and imputation strategy

3.1 Cleaning steps

1. Convert variable-specific sentinel codes to missing, and empty strings in text columns to missing. This step is stateless: it uses only the codebook, not statistics from the data, so it is safe to run before the split.
2. Keep "Not applicable" separate from "Missing" in the audit, because it comes from questionnaire skip logic rather than nonresponse. None of the selected features contains Not-applicable codes.
3. Apply domain validity rules. Ten training rows report a class enrolment of 0 (`qm22` = 0), which is impossible, so they become missing.
4. Recode the Yes/No items from 1/2 to 1/0.
5. Exclude redundant or unusable columns (`qe61y`, `qd26`, `qd27`, `qd11`, `qd12`, `qm3`, `qm4`) and all leakage columns (Section 5.5).

3.2 Missingness mechanism diagnosis

I use the three standard mechanisms:

- **MCAR (missing completely at random):** missingness is pure chance.
- **MAR (missing at random):** missingness depends on other observed variables.
- **MNAR (missing not at random):** missingness depends on the unobserved value itself or on something unmeasured that drives it.

Evidence, all computed on the training set:

- Little's MCAR test on the numeric features (`ses`, `qe62`, `qm22`) rejects MCAR ($\chi^2 = 329.5$, df = 9, $p < 0.001$).
- For every feature, missingness differs by country (chi-square $p < 0.001$, Cramér's V from 0.07 to 0.24).
- For every feature, the share of students below the minimum differs between missing and observed rows.

MAR and MNAR cannot be fully separated from observed data. The decisive evidence for MNAR is when the missing group does worse than every observed category of the same question.

Table 4: Missingness diagnosis per selected feature (training set, n = 50,347).

Feature	Miss. %	Country V	% below (miss. / obs.)	Interpretation	Mechanism
`ses`	3.9	0.133	79.7 / 62.4	Index not computed when asset items are unanswered; varies by country and age	MAR
`qe62`	0.3	0.068	77.0 / 63.0	Very rare; varies by country	MAR (negligible)
`qm22`	14.0	0.238	66.6 / 62.5	Mostly unit nonresponse: teacher questionnaire not linked for 9.6% of students, up to 27% in Chad	MAR
`qm5`	13.3	0.213	68.5 / 62.2	Same teacher block	MAR
`qd31`	3.5	0.136	55.6 / 63.3	Principal questionnaire not linked; country-driven	MAR
`qe621a`	10.9	0.163	68.5 / 62.4	Missing group does worse than "No" (65.1%) and "Yes" (60.0%): students who cannot say whether a parent reads likely come from lower-literacy homes	MNAR suspected
`qe621b`	11.6	0.171	68.1 / 62.4	Same pattern (missing 68.1 vs No 65.5, Yes 60.9)	MNAR suspected
`qe68`	6.5	0.153	75.7 / 62.2	Missing group does worse than both repeaters (65.6) and non-repeaters (57.9); weaker readers may skip self-completed items	MNAR suspected
`qe620`	6.4	0.175	82.6 / 61.7	Missing group does worse than every answer category (54.7 to 74.4); same reading-ability explanation	MNAR suspected

3.3 Imputation plan

Within-country imputation. Missingness depends on country for every feature, and feature distributions differ by country. A global median would pull a student in Chad towards values typical of Senegal. All imputers therefore compute their statistic per country on the training data, falling back to the global training statistic when a country has too few observed values. The choice of statistic follows each feature's shape (training set):

Table 5: Imputation rules.

Feature	Shape (train)	Imputation	Justification
`ses`	skew -0.09, kurtosis -0.56; mean 48.87, median 49.35	Country mean	Near-symmetric, so the mean is an unbiased centre and uses all information.
`qe62`	skew 0.64; mean 12.85, median 13	Country median	Moderate right skew from over-age students; the median is robust to the long tail.
`qm22`	skew 3.32, kurtosis 26.8; mean 39.3, median 35	Country median	Strong right skew (very large classes) would inflate a mean.
`qe621a`, `qe621b`, `qe68`	Binary	Country mode	Mean or median is undefined for categories.
`qm5`, `qd31`, `qe620`	Ordinal	Country mode	Keeps imputed values on valid category codes.

Missing-indicator flags. For every feature I add a binary column marking that the value was imputed. The missing rows differ from observed rows by 4 to 21 percentage points in the share below the minimum, so the fact that a value is missing is itself informative. The flags also cover the four MNAR-suspected features, where imputation alone would hide that signal.

Methods considered and rejected:

- **Listwise deletion** would drop 33.8% of students and bias the sample towards linked, complete schools.
- **KNN imputation** needs distances between mixed categorical codes, which are not meaningful, and costs $O(n^2)$ on 50,000 rows.
- **Iterative (MICE-style) imputation** is kept as a Phase 2 sensitivity check, not the main method.

XGBoost can also handle missing values natively, and I will compare this against the imputed version in Phase 2.

3.4 Outlier identification and treatment

Outlier rules apply to the three numeric features. The binary and ordinal features are bounded codes, already validated against the codebook. I use the z-score rule for the near-normal feature and the IQR rule for the skewed ones. The bounds below come from the full training set; in the pipeline they are re-learned on each training fold.

Table 6: Outlier rules and bounds (training set).

Feature	Rule	Bounds	Flagged	Observed range
`ses`	$\|Z\| > 3$ (near-normal)	$\mu \pm 3\sigma$ = [19.10, 78.64]	34 (0.07%)	27.4 to 79.6
`qe62`	1.5 × IQR (skewed)	Q1 = 12, Q3 = 14, IQR = 2: [9, 17]	540 (1.08%)	8 to 27 years
`qm22`	1.5 × IQR (skewed)	Q1 = 25, Q3 = 48, IQR = 23: [0, 82.5] (lower bound −9.5 floored at 0)	1,680 (3.88%)	4 to 352 students (after the 10 zero values are set to missing)

Treatment: winsorization (capping), not row deletion.

- **The extreme values are mostly real.** Over-age students and very large classes do exist in several PASEC countries, and they are part of the signal I want. Deleting them would remove exactly the students the model must learn about.
- **Deletion costs more than one value.** Removing a row for one extreme value also discards that student's valid answers on every other feature, and reduces N and statistical power.

Capping is applied only where it helps. The two models react differently to extreme values:

- **Logistic regression** fits one weight per feature, so a few extreme values (a class of 352 students) can pull that weight and distort the fit for everyone. For this model, values outside the bounds are set to the nearest bound. The bounds are learned on X_train only, through a custom transformer inside the pipeline.
- **XGBoost** splits on thresholds ("age above 14 or not"), so an extreme value cannot pull anything; it just falls on one side of a split. Capping would only remove detail, because it would make a 27-year-old identical to a 17-year-old. XGBoost therefore receives the original values.

No student and no value is deleted in either case. In Phase 2 I will report how many values were capped for logistic regression.

3.5 Statistical association strategy

Feature-target tests. The test for each feature is matched to its type against the binary target. Mutual information (MI) is computed for every feature as a common, model-free scale for ranking (Phase 2, Figure 3.3).

Table 7: Feature-target test matching matrix with preliminary training-set results (all $p < 0.001$ except where noted).

Feature	Type	Test against binary Y	Statistic	MI
`ses`	Continuous	Point-biserial r (confirmed with Mann-Whitney U)	$r = -0.216$	0.069
`qe62`	Discrete	Point-biserial r; Mann-Whitney U given skew	$r = 0.106$	0.008
`qm22`	Discrete	Point-biserial r; Mann-Whitney U given skew	$r = 0.010$ (p = 0.03)	0.015
`qm5`	Ordinal	Spearman ρ	$\rho = -0.087$	0.007
`qd31`	Ordinal	Spearman ρ	$\rho = 0.141$	0.014
`qe620`	Ordinal	Spearman ρ	$\rho = 0.157$	0.013
`qe621a`, `qe621b`	Binary	Chi-square, Cramér's V	V = 0.053, 0.045	<0.001
`qe68`	Binary	Chi-square, Cramér's V	V = 0.079	0.004
`ID_PAYS`	Nominal (14)	Chi-square, Cramér's V	V = 0.394	0.080

Feature-feature tests:

- numeric and ordinal pairs: Spearman ρ, since several features are skewed or ordinal;

- nominal pairs: Cramér's V;
- numeric against nominal: the ANOVA F test (eta squared).

Individually, the background features are only weakly associated with the label. Apart from country, no feature exceeds $|r| = 0.22$. I expect the model to rely on combinations of features, and I state openly that the targets may not be reached.

Multicollinearity protocol:

1. Pairwise screen: flag any pair with $|\rho| \geq 0.85$ (Spearman).
2. Compute the Variance Inflation Factor (VIF) on the encoded design matrix, and flag VIF > 10.
3. Decision rule for a flagged pair: drop the feature with the lower mutual information with Y. On a tie, drop the one with more missing data; on a further tie, drop the less interpretable one.

Result on the training set. The strongest pair is `ses` and `qd31` ($\rho = -0.558$), and no pair reaches 0.85. The highest VIF is 1.83 for the features and 2.56 for the country dummies, so all selected features are kept.

The rule was also applied to candidate columns I excluded. Birth year duplicates age exactly. Students present (`qm21`) and students enrolled (`qm22`) correlate at $\rho = 0.914$, and I keep `qm22` because it has fewer missing values (14.0% vs 14.8%) and measures class size directly. The raw household-asset items are not used alongside `ses`, since `ses` is built from them.

4 Encoding, scaling and transformation strategy

Encoding, matched to cardinality and order:

- **Binary** (`qe621a`, `qe621b`, `qe68`): 1 = Yes, 0 = No. A single column, no expansion.
- **Ordinal**, mapped to ordered integers that keep the rank without adding columns:
  - `qm5`: no training = 0 up to more than three years = 5;
  - `qd31`: city = 0, suburb = 1, large village = 2, small village = 3;
  - `qe620`: always = 0, often = 1, sometimes = 2, never = 3.
- **Nominal** `ID_PAYS` (14 categories): one-hot encoding, giving 14 columns, with unseen categories ignored. The cardinality is low, so one-hot adds little width and assumes no false order between countries. Target encoding is rejected: it computes category statistics from the label and creates a leakage risk that is unnecessary at this cardinality.

Scaling and transformation, matched to the algorithms.

Logistic regression (the interpretable baseline) uses an L2 penalty. The penalty treats all coefficients on the same scale, so the numeric and ordinal features are standardized (mean 0, SD 1) after imputation and capping.

- `ses` is near-normal, so StandardScaler applies directly.
- `qm22` is strongly right-skewed. A log transform, $\log(1+x)$, reduces its skew from 3.32 to -0.02 in the training data before standardization.
- `qe62` keeps its original scale after capping (skew 0.64).

MinMax scaling is not used: no model here needs inputs bounded in [0, 1] (no neural network, no Naive Bayes), and MinMax is sensitive to the extremes that capping only limits.

The main model is gradient-boosted trees (XGBoost). XGBoost builds trees one after another, and each new tree corrects the errors left by the previous ones. Trees split on thresholds, so they do not depend on feature scale or on monotonic transformations. For a fair comparison (the Model Development lecture: same conditions for every model), both models use the same features, the same cleaning and imputation, the same split and the same cross-validation folds. Only the steps that a model's mathematics requires differ: capping, the log transform and standardization are applied for logistic regression and not for XGBoost, since they would not change how a tree splits except to remove detail at the extremes.

PCA will be fitted in Phase 2 only as a diagnostic (scree plot, Figure 5.1), to show how much variance the encoded features share. It is not used for modelling: the design matrix has about 30 columns after encoding and indicators, and principal components would remove the interpretability that the prototype needs.

5 Split strategy and leakage prevention

5.1 Partitioning ratios

I use an **80/20 train/test split**: 50,347 training rows and 12,587 test rows. With N = 62,934, 20% still gives a test set of more than 12,000 students and about 4,600 negatives. That is enough for stable estimates per country and per subgroup, which the fairness checks need. The test set is used once, for final evaluation.

Hyperparameter tuning and threshold selection use **5-fold cross-validation inside the training set** (about 40,000 / 10,000 per fold), instead of a single validation split, so every training student is used for both fitting and validation.

5.2 Stratification

Splits are stratified on the combination of country and label (28 strata), not on the label alone. Country shares and per-country base rates differ widely (Figure 2), and the per-country review needs both preserved. The realised test split confirms this:

- the share below the minimum is 63.06% in train and 63.03% in test;
- country shares differ by at most 0.04 percentage points;
- per-country label rates differ by at most 0.35 percentage points.

5.3 Group-level separation

Students are clustered in schools. Students in the same class share identical teacher answers (`qm5`, `qm22`), and students in the same school share identical principal answers (`qd31`). If one school appeared in both train and test, the model could recognise the school instead of learning from the student, and test scores would be inflated. I therefore use `StratifiedGroupKFold`:

- groups are schools, identified by country plus `ID_ECOLE`;
- the test set is one of five folds;
- the training set has 2,639 schools and the test set 659, with zero overlap;
- the same grouping is used for the five tuning folds.

This also matches how the prototype would be used: on students from schools it has not seen.

5.4 Temporal alignment

PASEC2019 is a single cross-section collected in the 2018/19 school year. There is no time order between students, so a chronological split does not apply. Testing on a later PASEC round would be the temporal extension, but no later round is used in this project.

5.5 Leakage prevention architecture

Columns excluded from X:

- the 182 test items (`f1` to `f98`, `m1` to `m84`);
- all 10 plausible values: the maths values define the label, and reading is a parallel test outcome;
- the survey weights and jackknife variables (`rwgt0` to `rwgt90`, `W_ECOLE`, `JKZONE`, `JKREP`);
- all identifiers.

Pipeline order:

1. **Before the split, stateless steps only:**
   - sentinel conversion, which uses the codebook and not data statistics;
   - label construction, which uses a fixed published cutoff;
   - column selection.
2. **The split** (Sections 5.1 to 5.3).
3. **One scikit-learn** `Pipeline` holding a `ColumnTransformer` and the model. Every step that learns from data lives inside it:
   - shared steps for both models: within-country imputers with missing indicators (numeric, binary and ordinal features), ordinal mapping, and one-hot encoding of country;
   - logistic regression only: winsorizer, log transform for `qm22`, then StandardScaler;
   - XGBoost: no capping, log transform or scaling.

   Calling `fit` on X_train learns imputation values, capping bounds, scaling parameters and categories from training rows only.
4. **Cross-validation:** the whole Pipeline is passed to the CV search, so it is re-fitted inside every fold. The validation fold never contributes to the statistics used to transform it.
5. **Threshold tuning** on out-of-fold training predictions.
6. **Final evaluation:** the test set is transformed with the fitted pipeline and evaluated once.

Survey weights are not features. I will report test metrics both unweighted and weighted by `rwgt0`, so results can also be read at population level. Gender, the urban/rural flag and country are kept aside for the fairness audit.

6 Limitations

- The label is built from plausible values that PASEC generated using background variables. Feature-label association is therefore partly built in.
- 22.2% of students are borderline: their five plausible values do not all fall on the same side of the cutoff.
- Pooling 14 countries mixes different curricula. The per-country review is designed to show this, not hide it.
- Questionnaire answers are reported by 12-year-olds and may be inaccurate (for example, parental literacy).
- Two feature definitions are unverified: the urban/rural grouping of `qd31`, and `qe620` for students not tested in French.

References

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[5] R. Artstein and M. Poesio, "Inter-coder agreement for computational linguistics," *Computational Linguistics*, vol. 34, no. 4, pp. 555-596, 2008.

[6] R. J. A. Little, "A test of missing completely at random for multivariate data with missing values," *J. Amer. Statist. Assoc.*, vol. 83, no. 404, pp. 1198-1202, 1988.

[7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. KDD*, 2016.

[8] C. Huyen, *Designing Machine Learning Systems*, O'Reilly, 2022.

Appendix A. Variable inventory and baseline missingness, all 749 columns

Columns in file order. Type follows Section 2.2. Missing = cells that are blank in the file or hold a sentinel code (Section 2.3), counted on the full dataset before any processing (N = 62,934).

Column	Type	Role	Missing	%	Content
ID_TOT	Nominal	Identifier	0	0.00	Student identifier
ID_ECOLE	Nominal	Grouping key	0	0.00	School ID
ID_ELEVE	Nominal	Identifier	0	0.00	Student ID in a given school
ID_STRATE	Nominal	Identifier	0	0.00	Stratum identifier in a given country
ID_PAYS	Nominal	Selected feature	0	0.00	Country identifier
PAYS	Nominal	Identifier	0	0.00	Country name
LECT_PV1	Continuous	Excluded: reading score	0	0.00	First plausible value in reading
LECT_PV2	Continuous	Excluded: reading score	0	0.00	Second plausible value in reading
LECT_PV3	Continuous	Excluded: reading score	0	0.00	Third plausible value in reading
LECT_PV4	Continuous	Excluded: reading score	0	0.00	Fourth plausible value in reading
LECT_PV5	Continuous	Excluded: reading score	0	0.00	Fifth plausible value in reading
MATHS_PV1	Continuous	Label source	0	0.00	First plausible value in mathematics
MATHS_PV2	Continuous	Label source	0	0.00	Second plausible value in mathematics
MATHS_PV3	Continuous	Label source	0	0.00	Third plausible value in mathematics
MATHS_PV4	Continuous	Label source	0	0.00	Fourth plausible value in mathematics
MATHS_PV5	Continuous	Label source	0	0.00	Fifth plausible value in mathematics
f1	Nominal	Excluded: test item	33,000	52.44	Reading test item f1
f2	Nominal	Excluded: test item	32,818	52.15	Reading test item f2
f3	Nominal	Excluded: test item	33,151	52.68	Reading test item f3
f4	Nominal	Excluded: test item	33,102	52.60	Reading test item f4
f5	Nominal	Excluded: test item	33,033	52.49	Reading test item f5
f6	Nominal	Excluded: test item	33,313	52.93	Reading test item f6
f7	Nominal	Excluded: test item	33,309	52.93	Reading test item f7
f8	Nominal	Excluded: test item	33,143	52.66	Reading test item f8
f9	Nominal	Excluded: test item	33,266	52.86	Reading test item f9
f10	Nominal	Excluded: test item	33,332	52.96	Reading test item f10
f11	Nominal	Excluded: test item	35,382	56.22	Reading test item f11
f12	Nominal	Excluded: test item	33,458	53.16	Reading test item f12
f13	Nominal	Excluded: test item	33,631	53.44	Reading test item f13
f14	Nominal	Excluded: test item	33,512	53.25	Reading test item f14
f15	Nominal	Excluded: test item	33,560	53.33	Reading test item f15
f16	Nominal	Excluded: test item	34,134	54.24	Reading test item f16
f17	Nominal	Excluded: test item	33,963	53.97	Reading test item f17
f18	Nominal	Excluded: test item	33,421	53.10	Reading test item f18
f19	Nominal	Excluded: test item	34,323	54.54	Reading test item f19
f20	Nominal	Excluded: test item	34,131	54.23	Reading test item f20
f21	Nominal	Excluded: test item	33,448	53.15	Reading test item f21
f22	Nominal	Excluded: test item	33,657	53.48	Reading test item f22
f23	Nominal	Excluded: test item	34,323	54.54	Reading test item f23
f24	Nominal	Excluded: test item	34,755	55.22	Reading test item f24
f25	Nominal	Excluded: test item	32,493	51.63	Reading test item f25
f26	Nominal	Excluded: test item	33,532	53.28	Reading test item f26
f27	Nominal	Excluded: test item	32,538	51.70	Reading test item f27
f28	Nominal	Excluded: test item	32,709	51.97	Reading test item f28
f29	Nominal	Excluded: test item	33,211	52.77	Reading test item f29
f30	Nominal	Excluded: test item	32,943	52.35	Reading test item f30
f31	Nominal	Excluded: test item	33,243	52.82	Reading test item f31
f32	Nominal	Excluded: test item	32,888	52.26	Reading test item f32
f33	Nominal	Excluded: test item	33,256	52.84	Reading test item f33
f34	Nominal	Excluded: test item	32,955	52.36	Reading test item f34
f35	Nominal	Excluded: test item	33,156	52.68	Reading test item f35
f36	Nominal	Excluded: test item	32,867	52.22	Reading test item f36
f37	Nominal	Excluded: test item	32,925	52.32	Reading test item f37
f38	Nominal	Excluded: test item	32,704	51.97	Reading test item f38
f39	Nominal	Excluded: test item	32,701	51.96	Reading test item f39
f40	Nominal	Excluded: test item	32,611	51.82	Reading test item f40
f41	Nominal	Excluded: test item	32,672	51.91	Reading test item f41
f42	Nominal	Excluded: test item	33,218	52.78	Reading test item f42
f43	Nominal	Excluded: test item	33,004	52.44	Reading test item f43
f44	Nominal	Excluded: test item	32,922	52.31	Reading test item f44

Column	Type	Role	Missing	%	Content
f45	Nominal	Excluded: test item	33,096	52.59	Reading test item f45
f46	Nominal	Excluded: test item	32,805	52.13	Reading test item f46
f47	Nominal	Excluded: test item	33,352	53.00	Reading test item f47
f48	Nominal	Excluded: test item	34,116	54.21	Reading test item f48
f49	Nominal	Excluded: test item	32,816	52.14	Reading test item f49
f50	Nominal	Excluded: test item	33,020	52.47	Reading test item f50
f51	Nominal	Excluded: test item	33,123	52.63	Reading test item f51
f52	Nominal	Excluded: test item	33,477	53.19	Reading test item f52
f53	Nominal	Excluded: test item	33,349	52.99	Reading test item f53
f54	Nominal	Excluded: test item	33,125	52.63	Reading test item f54
f55	Nominal	Excluded: test item	33,370	53.02	Reading test item f55
f56	Nominal	Excluded: test item	32,841	52.18	Reading test item f56
f57	Nominal	Excluded: test item	33,215	52.78	Reading test item f57
f58	Nominal	Excluded: test item	33,323	52.95	Reading test item f58
f59	Nominal	Excluded: test item	33,137	52.65	Reading test item f59
f60	Nominal	Excluded: test item	32,925	52.32	Reading test item f60
f61	Nominal	Excluded: test item	33,977	53.99	Reading test item f61
f62	Nominal	Excluded: test item	33,361	53.01	Reading test item f62
f63	Nominal	Excluded: test item	33,201	52.76	Reading test item f63
f64	Nominal	Excluded: test item	33,134	52.65	Reading test item f64
f65	Nominal	Excluded: test item	33,024	52.47	Reading test item f65
f66	Nominal	Excluded: test item	33,073	52.55	Reading test item f66
f67	Nominal	Excluded: test item	33,454	53.16	Reading test item f67
f68	Nominal	Excluded: test item	33,298	52.91	Reading test item f68
f69	Nominal	Excluded: test item	33,231	52.80	Reading test item f69
f70	Nominal	Excluded: test item	33,239	52.82	Reading test item f70
f71	Nominal	Excluded: test item	34,090	54.17	Reading test item f71
f72	Nominal	Excluded: test item	32,997	52.43	Reading test item f72
f73	Nominal	Excluded: test item	33,682	53.52	Reading test item f73
f74	Nominal	Excluded: test item	33,695	53.54	Reading test item f74
f75	Nominal	Excluded: test item	33,198	52.75	Reading test item f75
f76	Nominal	Excluded: test item	33,232	52.80	Reading test item f76
f77	Nominal	Excluded: test item	33,545	53.30	Reading test item f77
f78	Nominal	Excluded: test item	34,315	54.53	Reading test item f78
f79	Nominal	Excluded: test item	33,659	53.48	Reading test item f79
f80	Nominal	Excluded: test item	33,806	53.72	Reading test item f80
f81	Nominal	Excluded: test item	33,789	53.69	Reading test item f81
f82	Nominal	Excluded: test item	33,824	53.75	Reading test item f82
f83	Nominal	Excluded: test item	34,186	54.32	Reading test item f83
f84	Nominal	Excluded: test item	33,638	53.45	Reading test item f84
f85	Nominal	Excluded: test item	33,909	53.88	Reading test item f85
f86	Nominal	Excluded: test item	33,983	54.00	Reading test item f86
f87	Nominal	Excluded: test item	34,409	54.67	Reading test item f87
f88	Nominal	Excluded: test item	34,108	54.20	Reading test item f88
f89	Nominal	Excluded: test item	33,948	53.94	Reading test item f89
f90	Nominal	Excluded: test item	34,007	54.04	Reading test item f90
f91	Nominal	Excluded: test item	33,659	53.48	Reading test item f91
f92	Nominal	Excluded: test item	33,855	53.79	Reading test item f92
f93	Nominal	Excluded: test item	34,177	54.31	Reading test item f93
f94	Nominal	Excluded: test item	34,149	54.26	Reading test item f94
f95	Nominal	Excluded: test item	34,060	54.12	Reading test item f95
f96	Nominal	Excluded: test item	34,224	54.38	Reading test item f96
f97	Nominal	Excluded: test item	33,672	53.50	Reading test item f97
f98	Nominal	Excluded: test item	33,794	53.70	Reading test item f98
m1	Nominal	Excluded: test item	33,285	52.89	Mathematics test item m1
m2	Nominal	Excluded: test item	33,964	53.97	Mathematics test item m2
m3	Nominal	Excluded: test item	32,689	51.94	Mathematics test item m3
m4	Nominal	Excluded: test item	33,923	53.90	Mathematics test item m4
m5	Nominal	Excluded: test item	32,865	52.22	Mathematics test item m5
m6	Nominal	Excluded: test item	32,939	52.34	Mathematics test item m6
m7	Nominal	Excluded: test item	32,773	52.08	Mathematics test item m7
m8	Nominal	Excluded: test item	32,595	51.79	Mathematics test item m8
m9	Nominal	Excluded: test item	32,574	51.76	Mathematics test item m9
m10	Nominal	Excluded: test item	33,034	52.49	Mathematics test item m10
m11	Nominal	Excluded: test item	33,145	52.67	Mathematics test item m11

Column	Type	Role	Missing	%	Content
m12	Nominal	Excluded: test item	32,931	52.33	Mathematics test item m12
m13	Nominal	Excluded: test item	33,124	52.63	Mathematics test item m13
m14	Nominal	Excluded: test item	32,924	52.32	Mathematics test item m14
m15	Nominal	Excluded: test item	32,916	52.30	Mathematics test item m15
m16	Nominal	Excluded: test item	33,161	52.69	Mathematics test item m16
m17	Nominal	Excluded: test item	33,309	52.93	Mathematics test item m17
m18	Nominal	Excluded: test item	35,235	55.99	Mathematics test item m18
m19	Nominal	Excluded: test item	33,337	52.97	Mathematics test item m19
m20	Nominal	Excluded: test item	33,334	52.97	Mathematics test item m20
m21	Nominal	Excluded: test item	33,616	53.41	Mathematics test item m21
m22	Nominal	Excluded: test item	32,353	51.41	Mathematics test item m22
m23	Nominal	Excluded: test item	32,156	51.09	Mathematics test item m23
m24	Nominal	Excluded: test item	32,226	51.21	Mathematics test item m24
m25	Nominal	Excluded: test item	32,608	51.81	Mathematics test item m25
m26	Nominal	Excluded: test item	32,199	51.16	Mathematics test item m26
m27	Nominal	Excluded: test item	32,700	51.96	Mathematics test item m27
m28	Nominal	Excluded: test item	32,237	51.22	Mathematics test item m28
m29	Nominal	Excluded: test item	32,142	51.07	Mathematics test item m29
m30	Nominal	Excluded: test item	32,239	51.23	Mathematics test item m30
m31	Nominal	Excluded: test item	32,421	51.52	Mathematics test item m31
m32	Nominal	Excluded: test item	32,177	51.13	Mathematics test item m32
m33	Nominal	Excluded: test item	32,766	52.06	Mathematics test item m33
m34	Nominal	Excluded: test item	32,641	51.87	Mathematics test item m34
m35	Nominal	Excluded: test item	32,357	51.41	Mathematics test item m35
m36	Nominal	Excluded: test item	32,258	51.26	Mathematics test item m36
m37	Nominal	Excluded: test item	32,537	51.70	Mathematics test item m37
m38	Nominal	Excluded: test item	32,460	51.58	Mathematics test item m38
m39	Nominal	Excluded: test item	32,304	51.33	Mathematics test item m39
m40	Nominal	Excluded: test item	32,442	51.55	Mathematics test item m40
m41	Nominal	Excluded: test item	32,626	51.84	Mathematics test item m41
m42	Nominal	Excluded: test item	32,564	51.74	Mathematics test item m42
m43	Nominal	Excluded: test item	32,496	51.64	Mathematics test item m43
m44	Nominal	Excluded: test item	32,461	51.58	Mathematics test item m44
m45	Nominal	Excluded: test item	32,631	51.85	Mathematics test item m45
m46	Nominal	Excluded: test item	33,736	53.61	Mathematics test item m46
m47	Nominal	Excluded: test item	32,536	51.70	Mathematics test item m47
m48	Nominal	Excluded: test item	32,358	51.42	Mathematics test item m48
m49	Nominal	Excluded: test item	32,444	51.55	Mathematics test item m49
m50	Nominal	Excluded: test item	32,982	52.41	Mathematics test item m50
m51	Nominal	Excluded: test item	32,849	52.20	Mathematics test item m51
m52	Nominal	Excluded: test item	32,522	51.68	Mathematics test item m52
m53	Nominal	Excluded: test item	32,585	51.78	Mathematics test item m53
m54	Nominal	Excluded: test item	32,720	51.99	Mathematics test item m54
m55	Nominal	Excluded: test item	33,003	52.44	Mathematics test item m55
m56	Nominal	Excluded: test item	32,678	51.92	Mathematics test item m56
m57	Nominal	Excluded: test item	32,638	51.86	Mathematics test item m57
m58	Nominal	Excluded: test item	32,719	51.99	Mathematics test item m58
m59	Nominal	Excluded: test item	32,586	51.78	Mathematics test item m59
m60	Nominal	Excluded: test item	32,759	52.05	Mathematics test item m60
m61	Nominal	Excluded: test item	33,715	53.57	Mathematics test item m61
m62	Nominal	Excluded: test item	33,736	53.61	Mathematics test item m62
m63	Nominal	Excluded: test item	33,733	53.60	Mathematics test item m63
m64	Nominal	Excluded: test item	32,740	52.02	Mathematics test item m64
m65	Nominal	Excluded: test item	35,148	55.85	Mathematics test item m65
m66	Nominal	Excluded: test item	33,174	52.71	Mathematics test item m66
m67	Nominal	Excluded: test item	33,674	53.51	Mathematics test item m67
m68	Nominal	Excluded: test item	33,140	52.66	Mathematics test item m68
m69	Nominal	Excluded: test item	32,663	51.90	Mathematics test item m69
m70	Nominal	Excluded: test item	32,952	52.36	Mathematics test item m70
m71	Nominal	Excluded: test item	33,129	52.64	Mathematics test item m71
m72	Nominal	Excluded: test item	32,937	52.34	Mathematics test item m72
m73	Nominal	Excluded: test item	32,983	52.41	Mathematics test item m73
m74	Nominal	Excluded: test item	33,122	52.63	Mathematics test item m74
m75	Nominal	Excluded: test item	33,201	52.76	Mathematics test item m75
m76	Nominal	Excluded: test item	33,108	52.61	Mathematics test item m76

Column	Type	Role	Missing	%	Content
m77	Nominal	Excluded: test item	33,090	52.58	Mathematics test item m77
m78	Nominal	Excluded: test item	33,374	53.03	Mathematics test item m78
m79	Nominal	Excluded: test item	33,110	52.61	Mathematics test item m79
m80	Nominal	Excluded: test item	32,980	52.40	Mathematics test item m80
m81	Nominal	Excluded: test item	33,611	53.41	Mathematics test item m81
m82	Nominal	Excluded: test item	33,263	52.85	Mathematics test item m82
m83	Nominal	Excluded: test item	33,504	53.24	Mathematics test item m83
m84	Nominal	Excluded: test item	33,432	53.12	Mathematics test item m84
LANGUE	Nominal	Not selected	0	0.00	Language
qe61d	Datetime	Not selected	18,110	28.78	Day of birth [1-31]
qe61m	Datetime	Not selected	18,110	28.78	Month of birth [1-12]
qe61y	Datetime	Not selected	196	0.31	Year of birth [1994-2011]
qe62	Discrete	Selected feature	196	0.31	Student age
qe63	Nominal	Fairness audit	17	0.03	Student gender
qe64a	Nominal	Not selected	4,378	6.96	Lives with mother
qe64b	Nominal	Not selected	5,451	8.66	Lives with father
qe64c	Nominal	Not selected	10,670	16.95	Lives with male guardian
qe64d	Nominal	Not selected	11,023	17.52	Lives with female guardian
qe64e	Nominal	Not selected	8,073	12.83	Lives with brothers/sisters
qe64f	Nominal	Not selected	9,037	14.36	Lives with grandparents/uncles/aunts
qe64g	Nominal	Not selected	10,550	16.76	Lives with other persons
qe65	Nominal	Not selected	3,919	6.23	Attended Quranic school before primary school
qe66	Nominal	Not selected	3,809	6.05	Attended pre-primary education (nursery/kindergarten...
qe67a	Nominal	Not selected	45,585	72.43	Public nursery school or preparatory class
qe67b	Nominal	Not selected	45,527	72.34	Private nursery school
qe67c	Nominal	Not selected	47,654	75.72	Community nursery school
qe68	Nominal	Selected feature	4,062	6.45	Have you ever repeated a grade?
qe69a	Nominal	Not selected	36,457	57.93	Repeated CP1
qe69b	Nominal	Not selected	36,562	58.10	Repeated CP2
qe69c	Nominal	Not selected	35,960	57.14	Repeated CE1
qe69d	Nominal	Not selected	38,612	61.35	Repeated CE2
qe69e	Nominal	Not selected	35,649	56.65	Repeated CM1
qe69f	Nominal	Not selected	36,057	57.29	Repeated CM2
qe610	Nominal	Not selected	4,879	7.75	Do you wear glasses?
qe611a	Nominal	Not selected	5,402	8.58	Difficulty seeing when reading the board
qe611b	Nominal	Not selected	8,001	12.71	Difficulty seeing when reading a book
qe612	Ordinal	Not selected	4,798	7.62	Do you eat at home before going to school?
qe613	Ordinal	Not selected	4,754	7.55	Are you hungry in class?
qe614	Nominal	Not selected	4,085	6.49	Do you usually eat at midday at school?
qe615	Nominal	Not selected	41,661	66.20	If yes, where do you get the meal?
qe616	Ordinal	Not selected	4,510	7.17	During the school year, do you do domestic work at h...
qe617	Ordinal	Not selected	4,222	6.71	When not at school, do you do agricultural work?
qe618	Ordinal	Not selected	4,155	6.60	When not at school, do you do small trade/commerce?
qe619	Ordinal	Not selected	4,241	6.74	When not at school, do you do manual/craft work or s...
qe620	Ordinal	Selected feature	4,066	6.46	Do you speak French at home?
qe621a	Nominal	Selected feature	6,926	11.01	Mother can read
qe621b	Nominal	Selected feature	7,358	11.69	Father can read
qe621c	Nominal	Not selected	18,271	29.03	Male guardian can read
qe621d	Nominal	Not selected	18,518	29.42	Female guardian can read
qe621e	Nominal	Not selected	9,208	14.63	Brothers/sisters can read
qe621f	Nominal	Not selected	13,917	22.11	Grandparents/uncles/aunts can read
qe621g	Nominal	Not selected	17,200	27.33	Other persons can read
qe622	Nominal	Not selected	3,744	5.95	Do you often receive homework from the teacher?
qe623	Nominal	Not selected	3,755	5.97	Are you helped to do your homework at home?
qe624	Ordinal	Not selected	28,009	44.51	If yes, how often?
qe625a	Nominal	Not selected	32,472	51.60	Mother helps with homework
qe625b	Nominal	Not selected	32,448	51.56	Father helps with homework
qe625c	Nominal	Not selected	36,068	57.31	Male guardian helps with homework
qe625d	Nominal	Not selected	36,169	57.47	Female guardian helps with homework
qe625e	Nominal	Not selected	31,178	49.54	Brothers/sisters help with homework
qe625f	Nominal	Not selected	35,012	55.63	Grandparents/uncles/aunts help with homework
qe625g	Nominal	Not selected	34,606	54.99	Other persons help with homework

Column	Type	Role	Missing	%	Content
qe626	Nominal	Not selected	7,044	11.19	Do you read at home?
qe627	Nominal	Not selected	3,061	4.86	Do you have your reading textbook in class?
qe628	Nominal	Not selected	20,439	32.48	If yes, can you take it home in the evening?
qe629	Nominal	Not selected	3,539	5.62	Do you have your mathematics textbook in class?
qe630	Nominal	Not selected	26,969	42.85	If yes, can you take it home in the evening?
qe631	Nominal	Not selected	3,891	6.18	Are there books in your home?
qe632	Ordinal	Not selected	36,490	57.98	How many books are in your home?
qe633	Nominal	Not selected	3,712	5.90	Is there electricity at home?
qe634	Nominal	Not selected	27,869	44.28	If yes, what is the main source of electricity?
qe635a	Nominal	Not selected	3,881	6.17	Household asset: Television
qe635b	Nominal	Not selected	5,248	8.34	Household asset: Computer/tablet
qe635c	Nominal	Not selected	4,547	7.23	Household asset: Radio
qe635d	Nominal	Not selected	5,041	8.01	Household asset: DVD player
qe635e	Nominal	Not selected	5,749	9.13	Household asset: Hi-Fi system
qe635f	Nominal	Not selected	4,574	7.27	Household asset: Mobile phone/smartphone
qe635g	Nominal	Not selected	4,902	7.79	Household asset: Refrigerator/freezer
qe635h	Nominal	Not selected	4,759	7.56	Household asset: Fan
qe635i	Nominal	Not selected	5,442	8.65	Household asset: Air conditioner
qe635j	Nominal	Not selected	5,191	8.25	Household asset: Stove/oven
qe635k	Nominal	Not selected	5,147	8.18	Household asset: Butane gas cooker
qe636a	Nominal	Not selected	3,719	5.91	Household asset: Table and chairs
qe636b	Nominal	Not selected	4,470	7.10	Household asset: Sewing machine
qe636c	Nominal	Not selected	4,418	7.02	Household asset: Electric iron
qe636d	Nominal	Not selected	4,415	7.02	Household asset: Car or truck
qe636e	Nominal	Not selected	4,965	7.89	Household asset: Tractor
qe636f	Nominal	Not selected	4,566	7.26	Household asset: Motorcycle/moped/scooter
qe636g	Nominal	Not selected	4,379	6.96	Household asset: Bicycle
qe636h	Nominal	Not selected	4,880	7.75	Household asset: Boat or dugout canoe
qe636i	Nominal	Not selected	5,033	8.00	Household asset: Cart
qe637a	Nominal	Not selected	4,388	6.97	House floor: Cement/brick
qe637b	Nominal	Not selected	6,258	9.94	House floor: Stone/tile
qe637c	Nominal	Not selected	6,104	9.70	House floor: Metal sheet/iron/zinc
qe637d	Nominal	Not selected	6,296	10.00	House floor: Wood
qe637e	Nominal	Not selected	6,391	10.16	House floor: Earth/mud (banco)
qe637f	Nominal	Not selected	6,807	10.82	House floor: Fabric/canvas/plastic (tent)
qe637g	Nominal	Not selected	6,982	11.09	House floor: Leaves/stems/bamboo
qe638	Nominal	Not selected	4,918	7.81	Are there latrines/toilets in your home?
qe639	Nominal	Not selected	7,358	11.69	Main source of water used at home
qe640	Ordinal	Not selected	4,227	6.72	I feel good at school (agree/disagree)
qe641	Ordinal	Not selected	4,190	6.66	I like reading (agree/disagree)
qe642	Ordinal	Not selected	4,149	6.59	I like mathematics (agree/disagree)
qe643	Ordinal	Not selected	4,113	6.54	With my teacher, I understand and learn well this ye...
qd1	Discrete	Not selected	3,796	6.03	Principal age [20-76]
qd2	Nominal	Not selected	2,394	3.80	Principal gender
qd3	Discrete	Not selected	4,097	6.51	Years as principal [0-47]
qd4	Discrete	Not selected	4,196	6.67	Years of teaching before becoming principal [0-40]
qd5	Nominal	Not selected	1,723	2.74	Received complementary pedagogical training as princ...
qd6	Discrete	Not selected	31,070	49.37	If yes, duration in days [0-90]
qd7	Nominal	Not selected	2,060	3.27	Received additional training in team management in p...
qd8	Discrete	Not selected	36,017	57.23	If yes, total duration in days [0-60]
qd9	Nominal	Not selected	2,191	3.48	Received additional training in school management in...
qd10	Discrete	Not selected	39,358	62.54	If yes, total duration in days [0-75]
qd11	Ordinal	Not selected	2,702	4.29	Highest level of education attained (country-specifi...
qd12	Nominal	Not selected	10,574	16.80	Highest professional teaching diploma (country-speci...
qd13	Nominal	Not selected	1,735	2.76	Intervenes on administrative matters for better clas...
qd14	Nominal	Not selected	1,729	2.75	Intervenes on pedagogical matters for better class m...
qd15a	Nominal	Not selected	4,570	7.26	Reason for intervention: inspection/supervision of t...
qd15b	Nominal	Not selected	4,119	6.54	Reason for intervention: pedagogical support to teac...
qd15c	Nominal	Not selected	4,924	7.82	Reason for intervention: student discipline support
qd16a	Nominal	Not selected	50,408	80.10	Reason for non-intervention: not necessary
qd16b	Nominal	Not selected	50,108	79.62	Reason for non-intervention: administrative workload...

Column	Type	Role	Missing	%	Content
qd16c	Nominal	Not selected	50,122	79.64	Reason for non-intervention: lack of training for pr...
qd16d	Nominal	Not selected	50,234	79.82	Reason for non-intervention: lack of motivation (sal...
qd16e	Nominal	Not selected	50,433	80.14	Reason for non-intervention: teacher refusal
qd17	Nominal	Not selected	3,242	5.15	School status
qd18	Nominal	Not selected	1,647	2.62	School has complete cycle (all grades)
qd19	Nominal	Not selected	2,997	4.76	School has standard single-level classes
qd20	Discrete	Not selected	21,679	34.45	If yes, number of standard classes [0-20]
qd21	Nominal	Not selected	1,822	2.90	School has multi-grade classes
qd22	Discrete	Not selected	49,227	78.22	If yes, number of multi-grade classes [0-20]
qd23	Nominal	Not selected	1,838	2.92	School has double-shift classes
qd24	Discrete	Not selected	55,126	87.59	If yes, number of double-shift classes [0-20]
qd25	Discrete	Not selected	2,366	3.76	Total number of teachers in school [1-20]
qd26	Datetime	Not selected	62,934	100.00	Date courses actually started this school year
qd26d	Datetime	Not selected	2,365	3.76	Day [1-31]
qd26m	Datetime	Not selected	2,365	3.76	Month [1-12]
qd26y	Datetime	Not selected	1,339	2.13	Year [2018]
qd27	Datetime	Not selected	62,934	100.00	Usual end-of-course date this school year
qd27d	Datetime	Not selected	3,395	5.39	Day [1-31]
qd27m	Datetime	Not selected	3,395	5.39	Month [1-12]
qd27y	Datetime	Not selected	1,339	2.13	Year [2019]
qd28	Nominal	Not selected	3,564	5.66	How are classes managed when a teacher is absent for...
qd29	Nominal	Not selected	1,674	2.66	Were there strikes during this school year?
qd30	Discrete	Not selected	50,487	80.22	If yes, number of strike days [0-92]
qd31	Ordinal	Selected feature	2,167	3.44	School is located in...
qd32a	Nominal	Not selected	2,815	4.47	Locality has: paved road
qd32b	Nominal	Not selected	2,501	3.97	Locality has: electricity
qd32c	Nominal	Not selected	3,029	4.81	Locality has: secondary school
qd32d	Nominal	Not selected	5,330	8.47	Locality has: health post/care centre
qd32e	Nominal	Not selected	3,717	5.91	Locality has: dispensary/health centre
qd32f	Nominal	Not selected	4,548	7.23	Locality has: hospital
qd32g	Nominal	Not selected	4,224	6.71	Locality has: bank
qd32h	Nominal	Not selected	3,484	5.54	Locality has: high school (lycee)
qd32i	Nominal	Not selected	4,652	7.39	Locality has: post office
qd32j	Nominal	Not selected	3,974	6.31	Locality has: savings bank/microcredit agency
qd32k	Nominal	Not selected	3,918	6.23	Locality has: cultural/social centre or library
qd33	Nominal	Not selected	1,871	2.97	School is accessible throughout the school year
qd34	Nominal	Not selected	55,992	88.97	If no, main reason for inaccessibility
qd35	Nominal	Not selected	1,867	2.97	School was inspected in the past two years
qd36	Discrete	Not selected	12,190	19.37	If yes, number of inspections [0-31]
qd37a	Nominal	Not selected	17,641	28.03	Purpose of inspection: teacher training
qd37b	Nominal	Not selected	14,426	22.92	Purpose of inspection: teacher supervision
qd37c	Nominal	Not selected	15,936	25.32	Purpose of inspection: administrative management
qd37d	Nominal	Not selected	14,480	23.01	Purpose of inspection: pedagogical management
qd37e	Nominal	Not selected	19,488	30.97	Purpose of inspection: human resources management
qd37f	Nominal	Not selected	18,156	28.85	Purpose of inspection: school environment management
qd37g	Nominal	Not selected	18,177	28.88	Purpose of inspection: facilities and infrastructure...
qd37h	Nominal	Not selected	18,936	30.09	Purpose of inspection: school educational project
qd38	Nominal	Not selected	11,499	18.27	Inspector provided advice/tools to resolve difficult...
qd39	Nominal	Not selected	1,743	2.77	School organises parent meetings
qd40a	Nominal	Not selected	61,561	97.82	Reason for no parent meetings: too many parents to c...
qd40b	Nominal	Not selected	61,394	97.55	Reason for no parent meetings: parents unavailable
qd40c	Nominal	Not selected	61,548	97.80	Reason for no parent meetings: difficulty reaching/c...
qd40d	Nominal	Not selected	61,334	97.46	Reason for no parent meetings: task under teachers' ...
qd41a	Discrete	Not selected	4,489	7.13	Number of parent meetings at principal's initiative ...
qd41b	Discrete	Not selected	8,914	14.16	Number of parent meetings at parents' association in...
qd41c	Discrete	Not selected	10,748	17.08	Number of parent meetings at school management commi...
qd42	Nominal	Not selected	2,444	3.88	Majority of parents attend these meetings
qd43a	Ordinal	Not selected	2,303	3.66	School informs parents about: overall school academi...
qd43b	Ordinal	Not selected	3,095	4.92	School informs parents about: school achievements (f...

Column	Type	Role	Missing	%	Content
qd43c	Ordinal	Not selected	2,541	4.04	School informs parents about: pedagogical objectives...
qd43d	Ordinal	Not selected	2,087	3.32	School informs parents about: school rules (frequenc...
qd43e	Ordinal	Not selected	2,554	4.06	School discusses with parents: concerns about school...
qd44a	Ordinal	Not selected	1,985	3.15	School informs parents about: child's learning progr...
qd44b	Ordinal	Not selected	2,014	3.20	School informs parents about: child's behaviour/well...
qd44c	Ordinal	Not selected	2,316	3.68	School discusses with parents: child's learning conc...
qd44d	Ordinal	Not selected	2,269	3.61	School encourages parents to help child with homewor...
qd45a	Ordinal	Not selected	4,562	7.25	Parent association (APE) exists
qd45b	Ordinal	Not selected	6,332	10.06	Parent-teacher association (APEE) exists
qd45c	Ordinal	Not selected	6,320	10.04	Mothers' educational association (AME) exists
qd45d	Ordinal	Not selected	6,974	11.08	School cooperative exists
qd45e	Ordinal	Not selected	4,809	7.64	School management committee (COGES) exists
qd46a	Nominal	Not selected	10,522	16.72	APEE/AME/COGES project: no projects
qd46b	Nominal	Not selected	9,234	14.67	APEE/AME/COGES project: extracurricular activities o...
qd46c	Nominal	Not selected	6,933	11.02	APEE/AME/COGES project: construction/rehabilitation
qd46d	Nominal	Not selected	7,162	11.38	APEE/AME/COGES project: purchase of educational equi...
qd46e	Nominal	Not selected	7,584	12.05	APEE/AME/COGES project: hiring/supporting teachers
qd46f	Nominal	Not selected	7,303	11.60	APEE/AME/COGES project: support for struggling stude...
qd47a	Nominal	Not selected	2,704	4.30	School participated in: NGO/cooperation agency partn...
qd47b	Nominal	Not selected	3,360	5.34	School participated in: twinning with a foreign scho...
qd47c	Nominal	Not selected	4,009	6.37	School participated in: pilot project
qd47d	Nominal	Not selected	3,194	5.08	School participated in: UNICEF programme
qd47e	Nominal	Not selected	3,503	5.57	School participated in: UNESCO programme
qd47f	Nominal	Not selected	3,339	5.31	School participated in: African Development Bank pro...
qd47g	Nominal	Not selected	3,713	5.90	School participated in: UNDP programme
qd47h	Nominal	Not selected	27,094	43.05	School participated in: other programme
qd48a	Nominal	Not selected	34,057	54.12	Partnership consists of: financial support
qd48b	Nominal	Not selected	33,585	53.37	Partnership consists of: construction/rehabilitation...
qd48c	Nominal	Not selected	34,015	54.05	Partnership consists of: infrastructure (classrooms/...
qd48d	Nominal	Not selected	33,280	52.88	Partnership consists of: vaccination campaign
qd48e	Nominal	Not selected	33,821	53.74	Partnership consists of: maintenance/repair of equip...
qd48f	Nominal	Not selected	33,237	52.81	Partnership consists of: provision of textbooks
qd48g	Nominal	Not selected	33,326	52.95	Partnership consists of: provision of school supplie...
qd48h	Nominal	Not selected	33,577	53.35	Partnership consists of: canteen/meal provision
qd48i	Nominal	Not selected	33,140	52.66	Partnership consists of: teacher pedagogical trainin...
qd48j	Nominal	Not selected	33,671	53.50	Partnership consists of: support for extracurricular...
qd48k	Nominal	Not selected	33,606	53.40	Partnership consists of: covering exam fees
qd48l	Nominal	Not selected	33,985	54.00	Partnership consists of: covering teacher salaries
qd48m	Nominal	Not selected	33,581	53.36	Partnership consists of: covering volunteer teacher ...
qd48n	Nominal	Not selected	33,413	53.09	Partnership consists of: tutoring for struggling stu...
qd49a	Nominal	Not selected	3,782	6.01	Community support: financial aid
qd49b	Nominal	Not selected	3,654	5.81	Community support: construction/rehabilitation/maint...
qd49c	Nominal	Not selected	4,109	6.53	Community support: infrastructure (classrooms/housin...
qd49d	Nominal	Not selected	3,487	5.54	Community support: vaccination campaign
qd49e	Nominal	Not selected	3,684	5.85	Community support: maintenance/repair of equipment
qd49f	Nominal	Not selected	3,660	5.82	Community support: provision of textbooks
qd49g	Nominal	Not selected	3,869	6.15	Community support: provision of school supplies
qd49h	Nominal	Not selected	3,686	5.86	Community support: canteen/meal provision
qd49i	Nominal	Not selected	4,030	6.40	Community support: teacher pedagogical training
qd49j	Nominal	Not selected	4,021	6.39	Community support: extracurricular activities
qd49k	Nominal	Not selected	3,627	5.76	Community support: covering exam fees
qd49l	Nominal	Not selected	4,110	6.53	Community support: covering teacher salaries

Column	Type	Role	Missing	%	Content
qd49m	Nominal	Not selected	3,912	6.22	Community support: covering volunteer teacher costs
qd49n	Nominal	Not selected	3,801	6.04	Community support: tutoring for struggling students
qd50a	Nominal	Not selected	3,135	4.98	Main reason for teacher absenteeism: health problems
qd50b	Nominal	Not selected	6,239	9.91	Main reason for teacher absenteeism: other income-ge...
qd50c	Nominal	Not selected	5,640	8.96	Main reason for teacher absenteeism: lack of motivat...
qd50d	Nominal	Not selected	5,433	8.63	Main reason for teacher absenteeism: long distance f...
qd50e	Nominal	Not selected	5,742	9.12	Main reason for teacher absenteeism: harsh teaching ...
qd50f	Nominal	Not selected	5,742	9.12	Main reason for teacher absenteeism: family visits
qd50g	Nominal	Not selected	3,761	5.98	Main reason for teacher absenteeism: social reasons
qd50h	Nominal	Not selected	4,695	7.46	Main reason for teacher absenteeism: travel to colle...
qd50i	Nominal	Not selected	5,343	8.49	Main reason for teacher absenteeism: strikes
qd50j	Nominal	Not selected	5,300	8.42	Main reason for teacher absenteeism: administrative ...
qd50k	Nominal	Not selected	5,601	8.90	Main reason for teacher absenteeism: insecurity/viol...
qd50l	Nominal	Not selected	5,382	8.55	Main reason for teacher absenteeism: political activ...
qd51	Nominal	Not selected	2,257	3.59	School has official French language curriculum docum...
qd52	Nominal	Not selected	17,840	28.35	If yes, school has all curriculum documents for the ...
qd53	Nominal	Not selected	2,701	4.29	School has official mathematics curriculum documents
qd54	Nominal	Not selected	19,207	30.52	If yes, school has all curriculum documents for the ...
qd55	Nominal	Not selected	3,297	5.24	School practices automatic grade promotion within su...
qd56	Nominal	Not selected	3,049	4.84	School practices automatic grade promotion between s...
qd57	Nominal	Not selected	2,031	3.23	Best students are officially recognised (honour boar...
qd58	Nominal	Not selected	1,942	3.09	School provides remedial support hours for weaker CP...
qd59	Ordinal	Not selected	37,652	59.83	If yes, average hours per week (CP2 remedial)
qd60	Nominal	Not selected	1,964	3.12	School provides remedial support hours for weaker CM...
qd61	Ordinal	Not selected	18,635	29.61	If yes, average hours per week (CM2 remedial)
qd62	Nominal	Not selected	19,021	30.22	During remedial hours, is there a homework-help syst...
qd63	Nominal	Not selected	1,799	2.86	Student enrolment register available
qd64	Nominal	Not selected	5,154	8.19	If yes, is it up to date?
qd65	Nominal	Not selected	1,815	2.88	Teacher attendance register exists
qd66	Nominal	Not selected	7,847	12.47	If yes, is it up to date?
qd67	Ordinal	Not selected	8,040	12.78	According to the register, how frequent are teacher ...
qd68	Ordinal	Not selected	7,851	12.47	According to the register, teacher absenteeism is...
qd69a	Discrete	Not selected	10,561	16.78	Average days absent per CP2 teacher in past 4 weeks ...
qd69b	Discrete	Not selected	9,315	14.80	Average days absent per CM2 teacher in past 4 weeks ...
qd70	Nominal	Not selected	2,080	3.31	School has defined a school project/action plan
qd71	Nominal	Not selected	30,026	47.71	If yes, is a document explaining the school project ...
qd72	Nominal	Not selected	1,396	2.22	School has a total operating budget for this school ...
qd73	Continuous	Not selected	29,631	47.08	If yes, total operating budget amount [0-1,000,000,0...
qd74	Nominal	Not selected	1,475	2.34	Annual school fees exist
qd75a	Continuous	Not selected	40,701	64.67	Registration fee amount [0-1,000,000,000]
qd75b	Continuous	Not selected	41,645	66.17	Tuition fee amount [0-1,000,000,000]
qd75c	Continuous	Not selected	46,602	74.05	Other fees amount [0-1,000,000,000]
qd76	Discrete	Not selected	2,317	3.68	Number of functional classrooms in school [0-43]
qd77a	Discrete	Not selected	41,529	65.99	Number of classrooms in cement [0-15]
qd77b	Discrete	Not selected	20,324	32.29	Number of classrooms in earth/mud (banco) [0-9]
qd77c	Discrete	Not selected	20,725	32.93	Number of classrooms in thatch [0-10]
qd78a	Nominal	Not selected	2,287	3.63	School has a separate principal’s office
qd78b	Nominal	Not selected	2,792	4.44	School has a secretariat
qd78c	Nominal	Not selected	2,534	4.03	School has a materials storage room
qd78d	Nominal	Not selected	2,520	4.00	School has a functioning library
qd78e	Nominal	Not selected	2,342	3.72	School has a computer room
qd78f	Nominal	Not selected	2,636	4.19	School has a separate staff room
qd78g	Nominal	Not selected	2,385	3.79	School has a playground/recreation yard

Column	Type	Role	Missing	%	Content
qd78h	Nominal	Not selected	2,664	4.23	School has an independent sports field
qd78i	Nominal	Not selected	2,266	3.60	School is completely fenced
qd78j	Nominal	Not selected	2,236	3.55	School has a first aid kit
qd78k	Nominal	Not selected	2,598	4.13	School has an infirmary
qd78l	Nominal	Not selected	2,333	3.71	School has housing for teacher(s)/principal
qd78m	Nominal	Not selected	2,804	4.46	School has running water
qd78n	Nominal	Not selected	2,755	4.38	School has another source of drinking water (well/bo...
qd78o	Nominal	Not selected	2,181	3.47	School has electricity
qd78p	Nominal	Not selected	2,215	3.52	School has a photocopier
qd78q	Nominal	Not selected	2,221	3.53	School has a computer
qd78r	Nominal	Not selected	2,354	3.74	School has internet connection
qd78s	Nominal	Not selected	2,274	3.61	School has a television
qd78t	Nominal	Not selected	2,255	3.58	School has a video cassette recorder or DVD player
qd79a	Nominal	Not selected	2,423	3.85	School has latrines
qd79b	Nominal	Not selected	2,571	4.09	School has flush toilets
qd80a	Discrete	Not selected	17,470	27.76	Number of boy toilets/latrines [0-8]
qd80b	Discrete	Not selected	17,650	28.05	Number of girl toilets/latrines [0-8]
qd80c	Discrete	Not selected	18,174	28.88	Number of male teacher toilets/latrines [0-6]
qd80d	Discrete	Not selected	18,927	30.07	Number of female teacher toilets/latrines [0-6]
qd81	Nominal	Not selected	12,874	20.46	Toilets/latrines are functional on survey day
qd82	Nominal	Not selected	12,968	20.61	Toilets/latrines have toilet paper
qd83	Nominal	Not selected	12,736	20.24	Toilets/latrines have water
qd84	Ordinal	Not selected	11,566	18.38	Frequency of toilet/latrine cleaning
qd85	Nominal	Not selected	2,789	4.43	School has a canteen
qd86a	Nominal	Not selected	49,072	77.97	Canteen type: free canteen
qd86b	Nominal	Not selected	50,838	80.78	Canteen type: subsidised/contribution canteen
ses	Continuous	Selected feature	2,501	3.97	Socio-economic status
INDICE_ INFRASTRUCTURES	Continuous	Not selected	1,803	2.86	School infrastructure index
INDICE_ IMPLI_COMMUNAU	Continuous	Not selected	1,514	2.41	Community involvement index
INDICE_ AMENAG_TERRI	Continuous	Not selected	1,876	2.98	Territorial planning index
JKREP	Nominal	Survey design	0	0.00	Dichomotous variable 1/0 in the JKZONE
JKZONE	Nominal	Survey design	0	0.00	Pseudo stratum
rwgt0	Continuous	Survey design	0	0.00	Student replicate weight 0
rwgt1	Continuous	Survey design	0	0.00	Student replicate weight 1
rwgt2	Continuous	Survey design	0	0.00	Student replicate weight 2
rwgt3	Continuous	Survey design	0	0.00	Student replicate weight 3
rwgt4	Continuous	Survey design	0	0.00	Student replicate weight 4
rwgt5	Continuous	Survey design	0	0.00	Student replicate weight 5
rwgt6	Continuous	Survey design	0	0.00	Student replicate weight 6
rwgt7	Continuous	Survey design	0	0.00	Student replicate weight 7
rwgt8	Continuous	Survey design	0	0.00	Student replicate weight 8
rwgt9	Continuous	Survey design	0	0.00	Student replicate weight 9
rwgt10	Continuous	Survey design	0	0.00	Student replicate weight 10
rwgt11	Continuous	Survey design	0	0.00	Student replicate weight 11
rwgt12	Continuous	Survey design	0	0.00	Student replicate weight 12
rwgt13	Continuous	Survey design	0	0.00	Student replicate weight 13
rwgt14	Continuous	Survey design	0	0.00	Student replicate weight 14
rwgt15	Continuous	Survey design	0	0.00	Student replicate weight 15
rwgt16	Continuous	Survey design	0	0.00	Student replicate weight 16
rwgt17	Continuous	Survey design	0	0.00	Student replicate weight 17
rwgt18	Continuous	Survey design	0	0.00	Student replicate weight 18
rwgt19	Continuous	Survey design	0	0.00	Student replicate weight 19
rwgt20	Continuous	Survey design	0	0.00	Student replicate weight 20
rwgt21	Continuous	Survey design	0	0.00	Student replicate weight 21
rwgt22	Continuous	Survey design	0	0.00	Student replicate weight 22
rwgt23	Continuous	Survey design	0	0.00	Student replicate weight 23
rwgt24	Continuous	Survey design	0	0.00	Student replicate weight 24
rwgt25	Continuous	Survey design	0	0.00	Student replicate weight 25
rwgt26	Continuous	Survey design	0	0.00	Student replicate weight 26
rwgt27	Continuous	Survey design	0	0.00	Student replicate weight 27
rwgt28	Continuous	Survey design	0	0.00	Student replicate weight 28

Column	Type	Role	Missing	%	Content
rwgt29	Continuous	Survey design	0	0.00	Student replicate weight 29
rwgt30	Continuous	Survey design	0	0.00	Student replicate weight 30
rwgt31	Continuous	Survey design	0	0.00	Student replicate weight 31
rwgt32	Continuous	Survey design	0	0.00	Student replicate weight 32
rwgt33	Continuous	Survey design	0	0.00	Student replicate weight 33
rwgt34	Continuous	Survey design	0	0.00	Student replicate weight 34
rwgt35	Continuous	Survey design	0	0.00	Student replicate weight 35
rwgt36	Continuous	Survey design	0	0.00	Student replicate weight 36
rwgt37	Continuous	Survey design	0	0.00	Student replicate weight 37
rwgt38	Continuous	Survey design	0	0.00	Student replicate weight 38
rwgt39	Continuous	Survey design	0	0.00	Student replicate weight 39
rwgt40	Continuous	Survey design	0	0.00	Student replicate weight 40
rwgt41	Continuous	Survey design	0	0.00	Student replicate weight 41
rwgt42	Continuous	Survey design	0	0.00	Student replicate weight 42
rwgt43	Continuous	Survey design	0	0.00	Student replicate weight 43
rwgt44	Continuous	Survey design	0	0.00	Student replicate weight 44
rwgt45	Continuous	Survey design	0	0.00	Student replicate weight 45
rwgt46	Continuous	Survey design	0	0.00	Student replicate weight 46
rwgt47	Continuous	Survey design	0	0.00	Student replicate weight 47
rwgt48	Continuous	Survey design	0	0.00	Student replicate weight 48
rwgt49	Continuous	Survey design	0	0.00	Student replicate weight 49
rwgt50	Continuous	Survey design	0	0.00	Student replicate weight 50
rwgt51	Continuous	Survey design	0	0.00	Student replicate weight 51
rwgt52	Continuous	Survey design	0	0.00	Student replicate weight 52
rwgt53	Continuous	Survey design	0	0.00	Student replicate weight 53
rwgt54	Continuous	Survey design	0	0.00	Student replicate weight 54
rwgt55	Continuous	Survey design	0	0.00	Student replicate weight 55
rwgt56	Continuous	Survey design	0	0.00	Student replicate weight 56
rwgt57	Continuous	Survey design	0	0.00	Student replicate weight 57
rwgt58	Continuous	Survey design	0	0.00	Student replicate weight 58
rwgt59	Continuous	Survey design	0	0.00	Student replicate weight 59
rwgt60	Continuous	Survey design	0	0.00	Student replicate weight 60
rwgt61	Continuous	Survey design	0	0.00	Student replicate weight 61
rwgt62	Continuous	Survey design	0	0.00	Student replicate weight 62
rwgt63	Continuous	Survey design	0	0.00	Student replicate weight 63
rwgt64	Continuous	Survey design	0	0.00	Student replicate weight 64
rwgt65	Continuous	Survey design	0	0.00	Student replicate weight 65
rwgt66	Continuous	Survey design	0	0.00	Student replicate weight 66
rwgt67	Continuous	Survey design	0	0.00	Student replicate weight 67
rwgt68	Continuous	Survey design	0	0.00	Student replicate weight 68
rwgt69	Continuous	Survey design	0	0.00	Student replicate weight 69
rwgt70	Continuous	Survey design	0	0.00	Student replicate weight 70
rwgt71	Continuous	Survey design	0	0.00	Student replicate weight 71
rwgt72	Continuous	Survey design	0	0.00	Student replicate weight 72
rwgt73	Continuous	Survey design	0	0.00	Student replicate weight 73
rwgt74	Continuous	Survey design	0	0.00	Student replicate weight 74
rwgt75	Continuous	Survey design	0	0.00	Student replicate weight 75
rwgt76	Continuous	Survey design	0	0.00	Student replicate weight 76
rwgt77	Continuous	Survey design	0	0.00	Student replicate weight 77
rwgt78	Continuous	Survey design	0	0.00	Student replicate weight 78
rwgt79	Continuous	Survey design	0	0.00	Student replicate weight 79
rwgt80	Continuous	Survey design	0	0.00	Student replicate weight 80
rwgt81	Continuous	Survey design	0	0.00	Student replicate weight 81
rwgt82	Continuous	Survey design	0	0.00	Student replicate weight 82
rwgt83	Continuous	Survey design	0	0.00	Student replicate weight 83
rwgt84	Continuous	Survey design	0	0.00	Student replicate weight 84
rwgt85	Continuous	Survey design	0	0.00	Student replicate weight 85
rwgt86	Continuous	Survey design	0	0.00	Student replicate weight 86
rwgt87	Continuous	Survey design	0	0.00	Student replicate weight 87
rwgt88	Continuous	Survey design	0	0.00	Student replicate weight 88
rwgt89	Continuous	Survey design	0	0.00	Student replicate weight 89
rwgt90	Continuous	Survey design	0	0.00	Student replicate weight 90
W_ECOLE	Continuous	Survey design	1,339	2.13	School weight
se8	Discrete	Not selected	55	0.09	Total students at start of primary school
ID_ENSEIGNANT	Nominal	Identifier	5,918	9.40	Teacher ID

Column	Type	Role	Missing	%	Content
classedp	Nominal	Not selected	8,589	13.65	Class tested at start of primary school
classefp	Nominal	Not selected	5,918	9.40	Class tested at end of primary school
qm1	Nominal	Not selected	7,412	11.78	Teacher gender
qm2	Discrete	Not selected	10,180	16.18	Teacher age [18-75]
qm3	Ordinal	Not selected	7,312	11.62	Highest level of education attained (country-specifi...
qm4	Nominal	Not selected	7,866	12.50	Highest professional teaching diploma (country-speci...
qm5	Ordinal	Selected feature	8,209	13.04	Duration of initial teacher training
qm6	Discrete	Not selected	15,357	24.40	Total duration of practical classroom training (mont...
qm7	Nominal	Not selected	7,251	11.52	Received in-service continuing professional training...
qm8	Discrete	Not selected	26,515	42.13	If yes, how many times? [a-b]
qm9a	Nominal	Not selected	28,399	45.13	Received additional training in French language dida...
qm9b	Nominal	Not selected	29,161	46.34	Received additional training in mathematics didactic...
qm9c	Nominal	Not selected	31,071	49.37	Received additional training in didactics of other s...
qm9d	Nominal	Not selected	29,886	47.49	Received additional training in pedagogy
qm9e	Nominal	Not selected	30,681	48.75	Received additional training in assessment of learni...
qm10	Discrete	Not selected	6,741	10.71	Years of teaching experience [0-61]
qm11ci	Nominal	Not selected	14,669	23.31	Has taught an introductory class (CI)
qm11cia	Discrete	Not selected	36,407	57.85	Number of times taught CI [0-44]
qm11cp	Nominal	Not selected	14,316	22.75	Has taught CP grade
qm11cpa	Discrete	Not selected	34,353	54.59	Number of times taught CP [0-40]
qm11ce1	Nominal	Not selected	13,028	20.70	Has taught CE1 grade
qm11ce1a	Discrete	Not selected	30,262	48.09	Number of times taught CE1 [0-37]
qm11ce2	Nominal	Not selected	16,966	26.96	Has taught CE2 grade
qm11ce2a	Discrete	Not selected	29,727	47.24	Number of times taught CE2 [0-40]
qm11cm1	Nominal	Not selected	11,353	18.04	Has taught CM1 grade
qm11cm1a	Discrete	Not selected	25,686	40.81	Number of times taught CM1 [0-33]
qm11cm2	Nominal	Not selected	11,177	17.76	Has taught CM2 grade
qm11cm2a	Discrete	Not selected	20,913	33.23	Number of times taught CM2 [0-43]
qm12	Discrete	Not selected	6,190	9.84	Number of years teaching at this school [a-b]
qm13	Nominal	Not selected	7,029	11.17	Current employment status
qm14	Discrete	Not selected	8,102	12.87	Number of days absent in past two months (excl. publ...
qm15a	Nominal	Not selected	22,783	36.20	Absent due to: health problems
qm15b	Nominal	Not selected	27,236	43.28	Absent due to: other income-generating activities
qm15c	Nominal	Not selected	26,859	42.68	Absent due to: lack of motivation
qm15d	Nominal	Not selected	27,027	42.94	Absent due to: long distance from home to school
qm15e	Nominal	Not selected	25,009	39.74	Absent due to: seminars/pedagogical days
qm15f	Nominal	Not selected	27,028	42.95	Absent due to: harsh teaching conditions
qm15g	Nominal	Not selected	25,920	41.19	Absent due to: social reasons (births/weddings/death...
qm15h	Nominal	Not selected	24,800	39.41	Absent due to: travel to collect salary
qm15i	Nominal	Not selected	26,771	42.54	Absent due to: strike
qm15j	Nominal	Not selected	26,319	41.82	Absent due to: administrative paperwork
qm15k	Nominal	Not selected	27,171	43.17	Absent due to: insecurity/violence/harassment at sch...
qm15l	Nominal	Not selected	27,255	43.31	Absent due to: political activities
qm15m	Nominal	Not selected	27,049	42.98	Absent due to: transport problems
qm15n	Nominal	Not selected	34,712	55.16	Absent due to: other reasons
qm16	Nominal	Not selected	7,905	12.56	Teacher receives salary regularly
qm17	Ordinal	Not selected	8,596	13.66	Frequency of salary payment
qm18a	Nominal	Not selected	12,532	19.91	Moonlights in: agriculture/livestock/fishing (past 3...
qm18b	Nominal	Not selected	17,276	27.45	Moonlights in: craft/small trade (past 3 months)
qm18c	Nominal	Not selected	18,301	29.08	Moonlights in: transport (past 3 months)
qm18d	Nominal	Not selected	16,680	26.50	Moonlights in: private teaching/tutoring (past 3 mon...
qm18e	Nominal	Not selected	17,990	28.59	Moonlights in: services (restaurant/phone booth etc....
qm19	Nominal	Not selected	8,499	13.50	Class organisation mode
qm20ci	Nominal	Not selected	27,231	43.27	Currently teaches CI grade
qm20cp	Nominal	Not selected	22,899	36.39	Currently teaches CP grade
qm20ce1	Nominal	Not selected	26,899	42.74	Currently teaches CE1 grade
qm20ce2	Nominal	Not selected	30,168	47.94	Currently teaches CE2 grade
qm20cm1	Nominal	Not selected	24,375	38.73	Currently teaches CM1 grade
qm20cm2	Nominal	Not selected	5,918	9.40	Currently teaches CM2 grade
qm21	Discrete	Not selected	9,321	14.81	Number of students present on last school day [0-150...
qm22	Discrete	Selected feature	8,788	13.96	Number of students enrolled in class

Column	Type	Role	Missing	%	Content
qm22a	Discrete	Not selected	7,457	11.85	Number of girls enrolled [0-182]
qm22b	Discrete	Not selected	7,599	12.07	Number of boys enrolled [0-250]
qm23a	Discrete	Not selected	8,015	12.74	Number of French textbooks in class [0-240]
qm23b	Discrete	Not selected	8,117	12.90	Number of mathematics textbooks in class [0-255]
qm24	Nominal	Not selected	7,826	12.44	Students allowed to take textbooks home
qm25a	Ordinal	Not selected	6,949	11.04	Proportion of students with pencil(s)/pen(s)
qm25b	Ordinal	Not selected	7,039	11.18	Proportion of students with notebook(s)
qm25c	Ordinal	Not selected	7,185	11.42	Proportion of students with a ruler
qm25d	Ordinal	Not selected	7,388	11.74	Proportion of students with a slate/chalkboard
qm25e	Ordinal	Not selected	7,377	11.72	Proportion of students with chalk
qm25f	Ordinal	Not selected	7,260	11.54	Proportion of students with a school bag
qm26a	Nominal	Not selected	7,392	11.75	Classroom has a teacher's desk
qm26b	Nominal	Not selected	7,298	11.60	Classroom has a teacher's chair
qm26c	Nominal	Not selected	7,726	12.28	Classroom has a mathematics pedagogical guide
qm26d	Nominal	Not selected	8,012	12.73	Classroom has a French pedagogical guide
qm26e	Nominal	Not selected	8,150	12.95	Classroom has a cupboard
qm26f	Nominal	Not selected	8,334	13.24	Classroom has a blackboard/whiteboard
qm26g	Nominal	Not selected	7,580	12.04	Classroom has chalk
qm26h	Nominal	Not selected	7,660	12.17	Classroom has ruler(s) for the board
qm26i	Nominal	Not selected	7,548	11.99	Classroom has set-square(s) for the board
qm26j	Nominal	Not selected	7,673	12.19	Classroom has compass(es) for the board
qm26k	Nominal	Not selected	8,059	12.81	Classroom has dictionaries
qm26l	Nominal	Not selected	8,092	12.86	Classroom has world map
qm26m	Nominal	Not selected	8,155	12.96	Classroom has map of Africa
qm26n	Nominal	Not selected	8,155	12.96	Classroom has globe
qm26o	Nominal	Not selected	7,882	12.52	Classroom has map of the country
qm26p	Nominal	Not selected	8,310	13.20	Classroom has bookshelves
qm26q	Nominal	Not selected	8,299	13.19	Classroom has a clock
qm26r	Nominal	Not selected	8,260	13.12	Classroom has a computer or tablet
qm27	Ordinal	Not selected	8,280	13.16	Classroom construction materials
qm28	Nominal	Not selected	7,784	12.37	Classroom has electricity
qm29	Discrete	Not selected	8,267	13.14	Number of seats in class [0-150]
qm30	Ordinal	Not selected	6,786	10.78	During reading lessons, teacher uses another languag...
qm31	Ordinal	Not selected	6,919	10.99	During mathematics lessons, teacher uses another lan...
qm32	Nominal	Not selected	7,696	12.23	Last year, teacher covered the entire official Frenc...
qm33	Ordinal	Not selected	38,459	61.11	If no, proportion of curriculum covered (French)
qm34	Nominal	Not selected	8,262	13.13	Last year, teacher covered the entire official mathe...
qm35	Ordinal	Not selected	41,575	66.06	If no, proportion of curriculum covered (mathematics...
qm36	Discrete	Not selected	13,199	20.97	Number of actual teaching hours per week [0-40]
qm37a	Nominal	Not selected	6,121	9.73	Gives most importance to: reading (French)
qm37b	Nominal	Not selected	6,125	9.73	Gives most importance to: text study (French)
qm37c	Nominal	Not selected	6,138	9.75	Gives most importance to: written expression (French...
qm37d	Nominal	Not selected	6,121	9.73	Gives most importance to: spelling/grammar/syntax (F...
qm37e	Nominal	Not selected	6,125	9.73	Gives most importance to: oral comprehension (French...
qm37f	Nominal	Not selected	6,138	9.75	Gives most importance to: oral expression (French)
qm38	Nominal	Not selected	16,832	26.75	Mathematics domain given the most time
qm39a	Nominal	Not selected	6,187	9.83	Mathematics time priority: counting/numeracy
qm39b	Nominal	Not selected	6,174	9.81	Mathematics time priority: arithmetic rules
qm39c	Nominal	Not selected	6,174	9.81	Mathematics time priority: geometric shapes and form...
qm39d	Nominal	Not selected	6,199	9.85	Mathematics time priority: mental arithmetic
qm39e	Nominal	Not selected	6,199	9.85	Mathematics time priority: applying geometry operati...
qm39f	Nominal	Not selected	6,199	9.85	Mathematics time priority: reasoning/problem-solving
qm40	Ordinal	Not selected	7,387	11.74	Frequency of pedagogical meetings with colleagues an...
qm41	Nominal	Not selected	8,102	12.87	Principal provides support in class
qm42a	Nominal	Not selected	14,376	22.84	Principal support: pedagogical domain
qm42b	Nominal	Not selected	15,729	24.99	Principal support: administrative domain
qm42c	Nominal	Not selected	17,183	27.30	Principal support: student discipline

Column	Type	Role	Missing	%	Content
qm43	Nominal	Not selected	7,113	11.30	Received a visit from a pedagogical supervisor (insp...
qm44a	Nominal	Not selected	24,639	39.15	Purpose of supervisor visit: teacher inspection
qm44b	Nominal	Not selected	31,072	49.37	Purpose of supervisor visit: meeting parents
qm44c	Nominal	Not selected	27,182	43.19	Purpose of supervisor visit: checking start of schoo...
qm44d	Nominal	Not selected	26,435	42.00	Purpose of supervisor visit: advising teachers on Fr...
qm44e	Nominal	Not selected	29,665	47.14	Purpose of supervisor visit: organising a training w...
qm44f	Nominal	Not selected	29,526	46.92	Purpose of supervisor visit: courtesy visit
qm44g	Nominal	Not selected	34,141	54.25	Purpose of supervisor visit: other reasons
qm45a	Nominal	Not selected	8,286	13.17	Teachers experience moral harassment in your school
qm45b	Nominal	Not selected	10,712	17.02	Teachers experience sexual harassment in your school
qm46a	Ordinal	Not selected	7,421	11.79	Satisfaction with salary level
qm46b	Ordinal	Not selected	7,607	12.09	Satisfaction with school curriculum
qm46c	Ordinal	Not selected	7,152	11.36	Satisfaction with school building quality
qm46d	Ordinal	Not selected	6,961	11.06	Satisfaction with classroom building quality
qm46e	Ordinal	Not selected	7,055	11.21	Satisfaction with availability of school supplies
qm46f	Ordinal	Not selected	7,013	11.14	Satisfaction with school management quality
qm46g	Ordinal	Not selected	7,165	11.38	Satisfaction with relationship with colleagues
qm46h	Ordinal	Not selected	7,088	11.26	Satisfaction with relationship with community
qm46i	Ordinal	Not selected	7,290	11.58	Satisfaction with regularity of salary payment
qm46j	Ordinal	Not selected	9,531	15.14	Satisfaction with promotion opportunities
qm46k	Ordinal	Not selected	8,197	13.02	Satisfaction with training opportunities
qm46l	Ordinal	Not selected	7,362	11.70	Satisfaction with school life
qm46m	Ordinal	Not selected	7,214	11.46	Satisfaction with relationships with students
qm46n	Ordinal	Not selected	7,172	11.40	Satisfaction with relationships with parents
qm46o	Ordinal	Not selected	7,452	11.84	Satisfaction with teacher authority in recent years
qm46p	Ordinal	Not selected	7,440	11.82	Satisfaction with educational team (staff and manage...
INDICE_ PERCEPT_MT	Continuous	Not selected	6,561	10.43	Teacher working conditions perception index
INDICE_ EQUIP_CLASSE	Continuous	Not selected	6,257	9.94	Classroom equipment index